

Relative Covariance

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Note: *Parts of this document may be AI-written. My general workflow is to write sections of the article myself, laying out the general points I want to hit on and going into detail where relevant. I then converse with an LLM to identify errors in the logic or math and opportunities to use clearer or more generalizable notation. I then have the LLM make the edits to reflect our proposed changes, including adding more detailed derivations.*

This article describes how the phenotypic covariance between two individuals i and j can be modeled by limited information we have about the pair. I derive formulas from the starting trait generative model and only introduce assumptions at the point the derivation requires it. I start with a simple trait architecture and iteratively add more complexity.

1 Additive genetics under random mating with independent environment

We begin with the simplest architecture: a trait determined by an additive genetic component and an independent environmental component, in a randomly mating population.

1.1 The phenotype model

For an individual i we write the (standardized) phenotype as

$$y_i = g_i + e_i, \tag{1}$$

where g_i is the genetic value and e_i the environmental deviation. We take y to be standardized across the population, $E[y] = 0$ and $\text{Var}[y] = 1$, and both components to be mean-centered, $E[g] = E[e] = 0$, with variances $\text{Var}[g] = V_A$ and $\text{Var}[e] = V_E$.

We then introduce the following three assumptions.

[A1] Additive genetic architecture

The genetic value is a linear combination of the individual's standardized genotypes at

the causal variants,

$$g_i = \sum_{k=1}^M \beta_k x_{ik},$$

where β_k is the per-standardized-allele causal effect of variant k , x_{ik} is the standardized genotype ($E[x_{ik}] = 0$, $\text{Var}[x_{ik}] = 1$), and M is the number of variants.

Commonly bundled into: the additive model; no dominance, no gene–gene interactions (i.e. GxG, epistasis), no gene–environment interactions (i.e., GxE).

[A2] Gene-environment independence

An individual’s environmental component is independent of their genetic component:

$$g \perp e$$

Commonly bundled into: no gene–environment correlation or interaction.

[A3] Independence of environment across individuals

The environmental component is independent across individuals (including relatives):

$$e_i \perp e_j \quad (i \neq j).$$

Commonly bundled into: no shared/common environment (C), no population stratification

Need to shift down subsequent Assumption identifiers

Because $g \perp e$, $\text{Cov}[g, e] = 0$ and the phenotypic variance decomposes as $\text{Var}[y] = V_A + V_E$. Narrow-sense heritability is typically defined as:

$$h^2 = \frac{V_A}{\text{Var}[y]}$$

Since $\text{Var}[y] = 1$, $h^2 = V_A$ under this model.

1.2 Phenotypic covariance is genotypic covariance

For a single fixed pair, y_i and y_j are two fixed values, so “their covariance” only has meaning as an average over an ensemble of pairs. The ensemble we use is the set of all pairs that share a given piece of *relatedness information* $\mathcal{I}_{ij}$ about the pair, and the object of interest is the conditional cross-product

$$C_{ij}(\mathcal{I}) := E[y_i y_j \mid \mathcal{I}_{ij}]. \tag{2}$$

If $E[y_i | \mathcal{I}] = 0$, then $C_{ij}(\mathcal{I}) = \text{Cov}[y_i, y_j | \mathcal{I}]$. For most information $\mathcal{I}$, like the realized relatedness or degree between two relatives, this equivalence is applicable. An exception is when the information we are modelling contains individuals' actual genotype information, for which $E[y_i | \text{genotypes}] = g_i \neq 0$, therefore the expected cross-product of two individuals is not equivalent to their covariance given the fixed information.

Using (1) and [A2], every cross term involving e vanishes in expectation, so the phenotypic cross-product reduces to the genetic one at every level:

$$C_{ij}(\mathcal{I}) = E[(g_i + e_i)(g_j + e_j) | \mathcal{I}] = E[g_i g_j | \mathcal{I}]. \quad (3)$$

The rest of the article evaluates (3) at four levels of resolution for $\mathcal{I}$.

1.3 Level 0 — the exact genetic cross-product

At the finest resolution we condition on the realized genotypes of i and j at the causal variants, for which we know the true causal effect sizes. Then g_i and g_j are fixed, and by [A1],

$$C_{ij} = E[y_i y_j | x_i, x_j, \beta] = g_i g_j = \sum_k \sum_{k'} \beta_k \beta_{k'} x_{ik} x_{jk'}. \quad (4)$$

Under our model and assumptions so far, this returns the true expected cross-product of two individuals' phenotypes. The genetic value covariance between two individuals is, fundamentally, an effect-size-weighted function of the genotypes they actually carry at causal sites. Notably, this formula does not require any knowledge of the relatedness of the two individuals.

1.4 Level 1: genome-wide allelic sharing

If we do not know the causal effect sizes nor which variants are causal, then we cannot apply (4). However, we can assume that genetic effects are independent of individuals' allelic sharing patterns.

[A3] Random, exchangeable causal effects

Causal effects are drawn independently and identically across variants, with

$$E[\beta_k] = 0, \quad \text{Var}[\beta_k] = \sigma_\beta^2, \quad \beta \perp \{x\},$$

and their distribution does not depend on whether two individuals share an allele for that variant.

Commonly bundled into: the infinitesimal model

What is the relevance of using the $\{x\}$ notation as opposed to just x ?

Do we actually have to assume that causal effects are drawn independently and identically across variants, or only that a pair's allelic sharing does not inform us of the effect sizes of the alleles they share/don't share. E.g., suppose that full-siblings not only shared

0.5 of the genome, but also the 0.5 they share for some reason tends to be in stronger-effect-size variants; then decomposing the expected 4-term product in (4) into a 2x2-term expected product would not be valid.

Taking the expectation of (4) over the effects, and using $\beta \perp \{x\}$ to factor:

$$\mathbb{E}[g_i g_j \mid x_i, x_j] = \sum_k \sum_{k'} \mathbb{E}[\beta_k \beta_{k'}] x_{ik} x_{jk'}.$$

Because the effects are i.i.d. with mean zero, $\mathbb{E}[\beta_k \beta_{k'}] = 0$ whenever $k \neq k'$ and σ_β^2 when $k = k'$; the entire off-diagonal ($k \neq k'$) collapses regardless of any correlation between the genotypes. What remains is

$$\mathbb{E}[g_i g_j \mid x_i, x_j] = \sigma_\beta^2 \sum_k x_{ik} x_{jk}.$$

The variance bookkeeping fixes σ_β^2 : since $\text{Var}[x_{ik}] = 1$, we have $V_A = \sum_k \mathbb{E}[\beta_k^2] \text{Var}[x_{ik}] = M\sigma_\beta^2$, hence $\sigma_\beta^2 = V_A/M$. Therefore

$$C_{ij}(x_i, x_j) = V_A \cdot \frac{1}{M} \sum_k x_{ik} x_{jk}. \quad (5)$$

The per-variant effect weighting of (4) has been replaced by the genome-wide genotype cross-product (i.e. allelic sharing), which we'll define by $S_{ij} = \frac{1}{M} \sum_k x_{ik} x_{jk}$. This can be thought of as an identity-by-state (IBS)-based measure of “relatedness”. Therefore, we can further simplify:

$$C_{ij}(S_{ij}) = V_A S_{ij}. \quad (6)$$

1.5 Level 2 — realized relatedness

The problem with calculating S_{ij} is that we may not have genotyped every causal variant that contributes to a trait. As long as some of the causal variants are not in perfect correlation with a genotyped variant, then the covariance of two individuals' phenotypes will not fully take into account the contributions of the ungenotyped causal variants. However, when individuals are related, their IBS-based relatedness from genotyped variants is informative of the IBS-based relatedness of ungenotyped variants. Therefore, the former can be used to capture the contributions of the latter.

When individuals inherit a genomic segment from a common ancestor, then all of the variants within that genomic segment will be identical-by-state, as long as we assume that de-novo mutations have not caused the inherited alleles to be different. These genomic segments and their variants are said to be identical-by-descent (IBD).

[A5] No mutation (IBD $\Rightarrow$ IBS)

Alleles that are identical-by-descent are also identical-by-state: descent is transmitted without mutation.

Commonly bundled into: the infinite-sites / no-recurrent-mutation assumption.

When two individuals' alleles are IBD, then they are always IBS; when they are not IBD, they can still be IBS. Therefore, we can decompose the formula for the IBS-based relatedness, S_{ij} , as the sum of the cross-products of IBD variants and the cross-products of non-IBD variants.

Write formula

For allele pairs that are IBD, then the expected cross-product in their allelic genotype is always 0.5 since they are identical and of variance 0.5. [I'm unsure about the math or how to argue this here](#) For allele pairs that are non-IBD, we have to make an assumption on how they are distributed.

[A6] Independence of non-IBD alleles

Any two alleles that are *not* IBD are independent draws from the population allele distribution.

Commonly bundled into: within an individual, no inbreeding (Hardy–Weinberg); between individuals, random mating, no population structure, and no assortative mating.

By the definition of independent draws, the expected cross-product of non-IBD alleles is therefore 0. For any particular variant, we can therefore summarize the expected correlation in a pairs' genotypes as a function of the number of alleles that are IBD between the pair

[Continue revising rest of document](#) We now evaluate the per-variant cross-product $E[x_{ik}x_{jk} | \mathcal{S}]$ in terms of IBD. It is cleanest to work at the level of individual alleles. At variant k (with population allele frequency p_k), let individual i carry the two allele indicators $a_{ik}^{(m)}, a_{ik}^{(p)} \in \{0, 1\}$ (maternal and paternal), each Bernoulli(p_k), so that the allele dosage is $G_{ik} = a_{ik}^{(m)} + a_{ik}^{(p)}$ and

$$x_{ik} = \frac{G_{ik} - 2p_k}{\sqrt{2p_k(1 - p_k)}}.$$

Two alleles are IBD if they descend from — and are copies of — the same ancestral allele. Let $IBD_{ij}(k) \in \{0, 1, 2\}$ denote the number of alleles individual i shares IBD with j at variant k .

Under [A6] the two alleles within an individual are independent, so $\text{Var}[G_{ik}] = 2p_k(1 - p_k)$ and the standardization indeed gives $\text{Var}[x_{ik}] = 1$. For the pair, conditioning on descent leaves the marginal allele distributions unchanged, so $E[x_{ik}] = 0$ and $E[x_{ik}x_{jk} | \mathcal{S}] = \text{Cov}[G_{ik}, G_{jk} | \mathcal{S}] / (2p_k(1 - p_k))$. Expanding the genotype covariance over the four ordered allele pairs,

$$\text{Cov}[G_{ik}, G_{jk} | \mathcal{S}] = \sum_{u \in \{m, p\}} \sum_{v \in \{m, p\}} \text{Cov}[a_{ik}^{(u)}, a_{jk}^{(v)}],$$

each term is $\text{Var}[a] = p_k(1 - p_k)$ if that allele pair is IBD (by [A5] it is the same copy, correlation 1) and 0 if it is not (by [A6], independent). If the pair shares $IBD_{ij}(k) = s$ alleles IBD, then exactly s of the four terms are nonzero, giving $\text{Cov}[G_{ik}, G_{jk} | s] = s p_k(1 - p_k)$ and hence the key per-variant identity

$$E[x_{ik}x_{jk} | IBD_{ij}(k) = s] = \frac{s}{2} \quad (s = 0, 1, 2). \quad (7)$$

That is, a variant contributes correlation 0, $\frac{1}{2}$, or 1 according to whether the pair shares 0, 1, or 2 alleles IBD there.

To turn the causal-variant average in (6) into a genome-wide quantity we need one more assumption.

[A4] Homogeneous genome sharing

The expected IBD sharing is the same at the causal variants as it is genome-wide; IBD is not enriched at particular loci.

Commonly bundled into: causal variants are representative of the genome with respect to relatedness.

Substituting (7) into (6) and applying [A4],

$$\frac{1}{M} \sum_k \mathbb{E}[x_{ik}x_{jk} \mid \mathcal{S}] = \frac{1}{M} \sum_k \frac{\text{IBD}_{ij}(k)}{2} \xrightarrow{\text{[A4]}} \pi_{ij},$$

where π_{ij} is the *realized relatedness* — the proportion of the pair’s genome that is IBD. We arrive at the central Level-2 result:

$$\boxed{\mathbb{E}[y_i y_j \mid \pi_{ij}] = \pi_{ij} V_A}. \quad (8)$$

The phenotypic covariance between two relatives is their realized fraction of shared genome times the additive variance. Because different pairs of the same nominal relationship share different realized fractions (full siblings scatter around $\frac{1}{2}$, and so on), (8) is expressed through the pair’s *actual* π_{ij} .

1.6 Level 3 — pedigree degree

Often we do not know π_{ij} but only the pair’s pedigree relationship (parent–offspring, full siblings, first cousins, ...).

[A7] Mendelian pedigree

The relationship degree determines the distribution of IBD sharing through Mendelian segregation, so that $\mathbb{E}[\pi_{ij} \mid \text{degree}]$ is the pedigree (expected) relatedness.

Commonly bundled into: a known pedigree with Mendelian inheritance.

Applying the tower property to (8) and averaging over the Mendelian distribution of π_{ij} given the degree d ,

$$\mathbb{E}[y_i y_j \mid d] = \mathbb{E}[\mathbb{E}[y_i y_j \mid \pi_{ij}] \mid d] = V_A \mathbb{E}[\pi_{ij} \mid d] = r_d V_A, \quad (9)$$

where $r_d = \mathbb{E}[\pi_{ij} \mid d] = 2\phi_{ij}$ is the expected relatedness (twice the kinship coefficient). This recovers the classical covariances between relatives:

Relationship	Expected relatedness r_d
Monozygotic twins	1
Parent–offspring	1/2
Full siblings	1/2
Half sibs / grandparent–grandchild / avuncular	1/4
First cousins	1/8
“Unrelated”	0

The step from (8) to (9) is exactly the realized-versus-expected distinction: Level 2 uses a pair’s true realized relatedness, whereas Level 3 replaces it with its pedigree expectation, and is therefore only as good as the pedigree and the (often large) sampling variance of π_{ij} around r_d allow — a gap that widens for distant relationships.

1.7 Summary of the tower

Each level conditions on coarser information than the one above and is its conditional expectation; the rightmost column lists the assumption(s) newly required to descend one step.

Level	Conditioned on	$E[y_i y_j \mid \cdot]$	New assumptions
0	causal genotypes	$g_i g_j$	[A1] (given [A2])
1	realized sharing $\mathcal{S}$	$V_A \frac{1}{M} \sum_k E[x_{ik} x_{jk} \mid \mathcal{S}]$	[A3]
2	realized relatedness π_{ij}	$\pi_{ij} V_A$	[A4], [A5], [A6]
3	pedigree degree d	$r_d V_A$	[A7]

Next architectures restart from Level 0 and re-introduce a subset of [A1]–[A7], relaxing others: e.g. a shared environment breaks [A2]; dominance/epistasis breaks [A1]; assortative mating and population structure break [A6] (correlating non-IBD alleles) and drive a wedge between realized and pedigree relatedness.